

Research Paper

GeoLLM: A specialized large language model framework for intelligent geotechnical design

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ABSTRACT

Large language models (LLMs) have achieved remarkable success in various industrial and research fields, enhancing work efficiency by assisting machines in comprehending human language. In geotechnical design where extensive repetitive cross-checking of design codes consumes considerable time and labour, the utilization of LLMs to enhance design procedures has not been explored before. The challenge is to ensure that LLMs accurately comprehend professional geotechnical information from text and execute mathematical calculations correctly. This study makes the first attempt at developing a specialized LLM framework, GeoLLM, integrated with an innovative prompt engineering strategy to extract professional information from text and enable accurate mathematical calculations. GeoLLM is applied to the design of single piles involving bearing capacity and settlement calculations. The results reveal that GeoLLM exhibits excellent performance in single pile cases. Additionally, compared with LLMs of varying architectures and sizes, commercial LLMs with over 100 billion parameters presented outstanding comprehensive capacities, while those with 1.8 ~ 72 billion parameters degraded relatively. These findings indicate the promising capacity of GeoLLM to address professional tasks in geotechnical design.

1. Introduction

Geotechnical design plays a critical role in ensuring the long-term safety of infrastructures and cost-effective management of construction within geotechnical engineering (Phoon et al., 2022). The design of infrastructures includes multiple repetitive tasks that necessitate manual cross-checking of design codes, leading to considerable time and labour consumption, and increasing the risk of design oversights (Otake & Honjo, 2022). Even a minor error in geotechnical design can result in catastrophic consequences, as evidenced by the leaning tower of Pisa (Maus & Nguyen, 2022), the Vajont Dam (Franci et al., 2020), and metro collapse accidents (Zhou et al., 2022). Therefore, an intelligent method that assists engineers in improving design efficiency and prevent such disasters in geotechnical design is in demand.

In recent years, artificial intelligence (AI) algorithms have attracted increasing interest in the field of geotechnical engineering. AI algorithms has been applied to solve various issues in the geotechnical field, such as material modelling for soil (Cao et al., 2021a; Zhang et al., 2021a, 2023; Shen et al., 2022; Li et al., 2022b, 2023b), frozen soil (Ren

et al., 2022; Li et al., 2023a, 2024) and rock (Wu et al., 2023), the structural stability of a slope (Soranzo et al., 2022), underground mine (Cao et al., 2021b) and tunnel (Zhao et al., 2023; Zhang et al., 2021b), as well as health monitoring of ground deformation (Yao et al., 2020; Yao & Song, 2023; Mai et al., 2023), shield tunnelling (Zhang et al., 2022; Chen et al., 2023b) and subgrade degradation (Liu et al., 2023a). Numerous AI algorithms have been applied and assessed for the feasibility and efficiency, including feedforward neural networks (FNN) (Zhao & Wu, 1994; Li et al., 2022a), convolutional neural networks (CNN) (Tong et al., 2017; Yang et al., 2021), recurrent neural network (RNN) (Zhu et al., 1998; Zhang et al., 2021a, 2023), graph neural networks (GNN) (Mei et al., 2022; Choi & Kumar, 2024), transformer structure (Chen et al., 2023a; Ma et al., 2024), Bayesian inference (Rackwitz, 2000; Tang et al., 2023a), random forest (Dong et al., 2011; Atangana Njock et al., 2022), XGBoost (Liu et al., 2019; Lyu & Yin, 2023), etc. Table 1 provides a summary of current AI algorithms widely adopted across geotechnical engineering, confirming AI algorithms as promising and reliable alternatives for various geotechnical tasks.

However, the majority of these AI algorithms have focused on

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0266-352X/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1
Synopsis of AI algorithms in geotechnical engineering.

Algorithms (year)	Core feature	Data format	Applied area
FNN (1994)	Fully connected layer Forward propagation	Numerical data	Material modelling Structural stability
RNN (1998)	Sequence dependency Hidden State	Numerical data	Material modelling Structural stability Subgrade degradation
Bayesian (2000)	Probability model Bayes theorem	Numerical data	Health monitoring Material modelling
random forest (2011)	Decision tree Importance evaluation	Numerical data	Structural stability
CNN (2017)	Shared Weights Local receptive field	Image and numerical data	Particle identification Material modelling Structural stability
XGBoost (2019)	Gradient boosting Importance evaluation	Numerical data	Structural stability
GNN (2022)	Message Notes Aggregation Functions	Numerical data	Material modelling
Transformer (2023)	Multi-Head & self- Attention mechanism	Numerical data	Health monitoring

evaluating or predicting well-structured infrastructures using given sequential data, images or numerical data during construction or operational phases, with limited relevance to geotechnical design. The geotechnical design demands the comprehensively understanding of the intricate professional information from design texts and precise execution of mathematical calculations, posing a persistent challenge for current AI algorithms.

Large language models (LLMs) have recently attracted significant attention in both industry and academia due to their remarkable success in text comprehension and generating human-like responses. These models are pre-trained on extensive internet data and comprise billions of parameters, enabling powerful capabilities in general-purpose language understanding and generation. Driven by transformer architecture and extensive text datasets, LLMs exhibit remarkable proficiency in comprehending complex language, conducting text analysis, and generating creative productions (Yang et al., 2023b). This technological process has transformed operational methods and paradigms across multiple domains, including chatbot (Kocou et al., 2023) and construction (Zheng et al., 2022). LLMs also hold promise in aiding the resolution of geotechnical engineering problems, such as slope stability, seismic risk, and numerical simulation (Wu et al., 2024). Geotechnical problems involve extensive professional expertise such as soil conditions, load considerations, and optimal structural configuration, along with mathematical calculations including bearing capacity assessments and settlement estimation. Although current LLMs exhibit robust text comprehension capacities, they still struggle with the complex geotechnical knowledge and mathematical calculations.

This study develops a novel and specialized LLM framework (GeoLLM) for geotechnical design. The innovations of GeoLLM involve a hybrid prompt engineering strategy to facilitate LLMs in understanding and extracting the professional geotechnical knowledge, and a mathematical toolset module designed for accurate calculations. This study is organized as follows: section 1 explains the significance of an intelligent method for geotechnical design, followed by a brief review of LLMs in Section 2. Section 3 provides the general framework of proposed GeoLLM for intelligent geotechnical design. Subsequently, the GeoLLM is instantiated and validated through a single pile dataset in section 4. The results and discussions conduct a comprehensive analysis of the performance of single pile dataset, the impact of size and architecture of

LLMs, computational effort, as well as potential limitations and future work.

2. Literature review of LLMs

Due to their exceptional capabilities, LLMs, the game-changers of AI, are serving as the cornerstone of natural language processing (Borzunov et al., 2022). Their adeptness in text comprehension and human-like response stems from the revolutionary transformer architecture (Kalyan, 2024). Transformer architecture is primarily composed of text embedding, positional encoding, multiple stacked encoders and decoders. This architecture enables LLMs to process the geotechnical text input sequentially using encoders and generate words autoregressively with decoders, as shown in Fig. 1. Text embedding converts geotechnical texts as sequential text vectors to represent their semantic information (Vaswani et al., 2017). These text vectors facilitate computers to understand and process human language. (Neelakantan et al., 2022). Positional encoding captures the positional information of words within the geotechnical text as fixed-dimensional vectors, which are then integrated with text vectors through concatenation or addition to create final embedding vectors representing geotechnical texts (Golovneva et al., 2024). Each encoder mainly involves multi-head attention and residual connection blocks. The multi-head attention block consists of multiple attention sub-blocks, which focus on different aspects of the semantic relationships between words. The multi-head attention block eventually converts the embedding vectors as attention vectors through a series of normalization and residual connection blocks, enabling the transformer to capture a variety of linguistic features and dependencies simultaneously (Cordonnier et al., 2020; Xie et al., 2023). With a similar architecture with encoder, the decoder will translate the attention vectors into words and generate an output sequence text. The excellent performance of LLMs is further complemented by the wealth of available data (Kalyan, 2024). The exponential growth of training data fuels the refinement of robust parameters, advances in generalization, and improvement in performance (Bahri et al., 2021).

LLMs can be broadly categorized into 3 types based on variations in their transformer architecture, including encoder-only, encoder-decoder and decoder-only LLMs. Encoder-only LLMs consists solely of transformer encoders, such as BERT (Devlin et al., 2019). Encoder-decoder LLMs utilize both encoders and decoders, such as T5 (Raffel et al., 2019) and BART (Lewis et al., 2020). Decoder-only LLMs employ decoders only, such as ChatGLM (Zeng et al., 2023), GPT-3.5 (Ouyang et al., 2022), GPT-4 (Achiam et al., 2023), Baichuan (Yang et al., 2023a), Qwen (Bai et al., 2023), LLaMA (Touvron et al., 2023) and Gemini (Anil et al., 2023). Due to the superior generative performance capacity, decoder-only LLMs have been mainstream models leveraged as decision-support toolsets across diverse domains. For examples, LLMs improve the efficiency of legal services serving as expert assistants, achieving an accuracy score of 0.92 (Nay et al., 2024). In engineering, LLMs are also utilized for project analysis, with accuracy rates exceeding 0.90 (Bai et al., 2024; Smetana et al., 2024).

LLMs can be easily deployed via local, cloud or edge devices for engineering application. Local deployment, which directly runs LLMs on local computers, is characterized by strong stability, customization, and controllability. However, this approach entails powerful and expensive hardware resources. Therefore, local deployment is suitable for LLMs with 14 billion parameters or less. In comparison, cloud deployment can achieve long-text inferences via the rapid employment of 100 billion-level LLMs by consumer-grade computers. Specifically, LLMs can run on cloud platforms through an application programming interface (API). Lastly, edge deployment places LLMs on base stations close to users, contributing to mobile applications but requiring edge devices with robust processing and memory capabilities (Lin et al., 2023).

However, LLMs are susceptible to generating factually incorrect, nonsensical or irrelevant responses, which is also called hallucinations. Prompt engineering is a supplemental principle for employing LLMs in

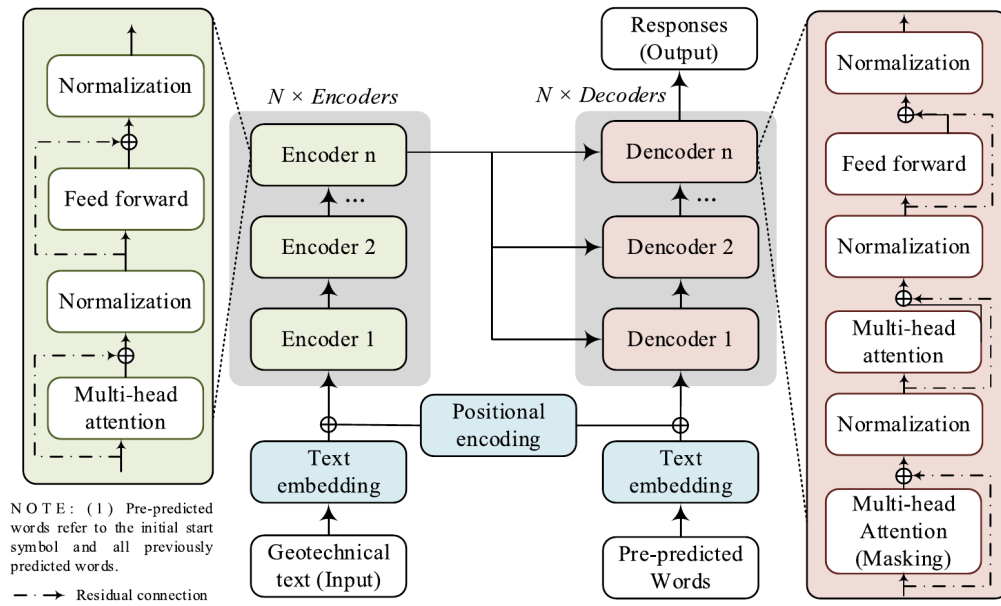


Fig. 1. Encode-decode transformer architecture.

different research topics (Marvin et al., 2024). A prompt in this study is a pre-designed text describing geotechnical design tasks to be executed by LLMs. Thus, prompt engineering guides LLMs to produce specific responses according to well-crafted prompts, which is crucial for LLMs to understand geotechnical texts and generate high-quality responses (Jessen et al., 2023). Table 2 presents the principles and

Table 2
Common prompt engineering techniques.

Techniques	Principle	Recommendation
Zero-shot (Wang et al., 2019)	It directly instructs LLMs to perform a geotechnical task without examples.	Simple and general tasks, such as geotechnical report summary.
Few-shot (Brown et al., 2020)	It provides demonstrations or examples to steer LLMs.	Tasks that need in-content learning for guiding LLMs' generation.
Retrieval augmented generation (RAG) (Lewis et al., 2020)	It uses retrieval components to search similar information and generate prompts.	Knowledge-intensive tasks.
Generated knowledge (Liu et al., 2021)	Generate knowledge from LLMs and provide this knowledge as additional information.	rapid-change or fast-updating tasks.
Chain-of-thought (COT) (Wei et al., 2022)	It takes another task's thought processes or intermediate reasoning steps as examples.	Multi-step reasoning tasks.
Self-consistency (Wang et al., 2022)	It samples multiple, diverse reasoning processes through few-shot CoT.	Tasks with COT-unsolvable reasoning processes.
ReAct (Yao et al., 2022)	It integrates COT reasoning traces and specific actions, such as searching.	Tasks with COT-unsolvable reasoning processes.
Tree-of-thought (TOT) (Yao et al., 2023)	It uses a tree structure with varied intermediate thoughts to assist LLMs in systematic lookahead and backtracking.	Tasks with COT-unsolvable reasoning processes
Directional stimulus prompting (Li et al., 2023c)	It uses repeated important words as hints.	Simple tasks but need a little directional instruction.
Graph prompting (Liu et al., 2023b)	It uses a graph structure as prompts.	Specially for graph-related tasks.
Prompt chain (Marvin et al., 2024)	It breaks complex tasks into step-by-step sub-tasks and uses the results of each sub-tasks as prompts.	Very complex multi-step reasoning tasks.

recommendation fields of common prompt engineering techniques.

The fine-tuning techniques can also facilitate LLMs to understand specialized texts and addressing hallucinations, by adjusting partial or all model parameters of LLMs (Ding et al., 2023). These methods can be categorized into supervised fine tuning (SFT), reinforcement learning with human feedback (RLHF), and reinforcement learning with AI feedback (RLAHF), according to the training strategies. SFT uses annotated data to refine LLMs through supervised learning (Lester et al., 2021). RLHF incorporates artificial feedback to update LLMs by maximizing rewards in alignment with human preferences (Li et al., 2023d), while RLAHF utilizes AI-generated feedback instead of artificial feedback (Lee et al., 2023). The fine-tuning methods can be also categorized into prompt tuning, prefix tuning, low-rank Adaptation (LoRA), and adaptor, based on architectural modifications. Prompt tuning updates partial parameters of LLMs within text embedding to explicitly embed prompt engineering into refined weights (Lester et al., 2021). Prefix tuning further utilizes tunable prefix vectors added to text vectors, implicitly incorporating prompt engineering techniques (Li & Liang, 2021). LoRA introduces low-rank matrix decompositions to update LLMs' parameters (Hu et al., 2021), while adaptor technique inserts small additional networks into LLMs for tuning (Hu et al., 2023).

Both prompt engineering and fine-tuning techniques have unique advantages and applicability. Prompt engineering can assist LLMs in comprehending geotechnical design tasks via well-crafted prompts, avoiding model retraining and enabling general versatility across diverse tasks using varying prompts. However, this method is usually limited by the maximum input-word count of commercial LLMs. On the other hand, fine-tuning techniques retrain LLMs with well-prepared geotechnical datasets, embedding geotechnical knowledge as weights in LLMs. However, retrained LLMs are tailored solely for specific tasks associated with the training datasets, with poor generalization. Furthermore, fine-tuning LLMs demands substantial computing resources and expenses which are impractical for individual engineers and small-scale industries.

Although prompt engineering and fine-tuning techniques can improve text comprehension of LLMs, LLMs still struggle with mathematical calculation and reasoning, as they are specifically designed for language generation and comprehension (Saba, 2024). Even the advanced LLMs presented high error rates in mathematical calculations based on professional datasets, such as MMLU dataset full of general reasoning tasks (Hendrycks et al., 2020), GSM8K dataset consisting of

middle-school math problems (Cobbe et al., 2021), and MATH dataset covering math problems from grades 4 to 12 (Hendrycks et al., 2021). One of the most powerful LLM, GPT-4o, achieved only 88.7 % accuracy on the MMLU dataset, 90.5 % accuracy on the GSM8K dataset and 76.6 % accuracy on the MATH dataset (OpenAI, 2024), respectively. These findings uncover that LLMs often exhibit declined precision and reliability in complex calculation and poor robustness across various reasoning tasks.

Based on the above-mentioned characteristics of LLMs, a specialized geotechnical LLM based on prompting engineering and mathematical toolsets is in demand, which is more applicable and can be easily extended to other geotechnical design tasks.

3. Proposed framework – GeoLLM

GeoLLM integrates LLMs with a prompt engineering strategy for text comprehension, along with a mathematical toolset module for computation, as shown in Fig. 2. Given a professional text related to geotechnical design, the LLM module is responsible for extracting key information as structured data, consisting of key-value pairs, using well-designed prompts. The key-value pair store properties as “key”, such as Poisson’s ratio and unit weight, and store the data of properties as “value”. The mathematical toolset module leverages the structured data for logical decision-making, calculation, and validation according to related design codes. Afterwards, solutions for geotechnical design are stored in readable files and in feedback to engineers.

3.1. LLM module

Considering the long complex texts in geotechnical design, the LLM module adopts pre-trained LLMs with over 100 billion parameters to provide rapid and accurate responses via cloud deployment.

The proposed hybrid prompt engineering strategy for the LLM module is derived from the techniques outlined in Table 2. The intelligent design of geotechnical engineering necessitates that LLMs should accurately extract professional information, including design requirements, parameters and codes, from geotechnical texts. Typically, original LLMs that have not been trained by geotechnical data struggle to extract this information. Therefore, few-shot prompt engineering technique is essential to incorporate typical examples into prompts for optimal performance. This technique integrates geotechnical instances as professional knowledge to assist LLMs in comprehending geotechnical design. Additionally, COT prompt engineering technique is leveraged to enhance the reasoning capacities of LLMs in geotechnical design.

This technique integrates incremental reasoning and definitive solutions to evoke more structured and insightful responses from LLMs. Furthermore, as few-shot and CoT techniques may not cover all design scenarios, the prior knowledge prompt engineering technique is used to bridge these gaps. This technique supplements unusual principles with explicit instructions, ensuring the efficiency of LLMs in generating high-quality responses. As the machine-readable data is essential for the mathematical toolset module in bearing capacity and settlement calculations, the response formatting prompt engineering technique is adopted to standardize the responses of LLMs into a dictionary format. Finally, these four prompt techniques are integrated as the hybrid prompt engineering strategy into the GeoLLM for geotechnical design. Every prompt in the proposed hybrid engineering would take its effects in helping LLMs to learn the geotechnical information. In our experiments, using response formatting or other single prompt only is not sufficient for advanced LLMs to accurately extract the geotechnical information from the input. Other techniques in Table 2 developed from the above four techniques with comparable impacts, are currently not under consideration. This hybrid strategy is well-suited for GeoLLM in geotechnical design, offering cost-effective and high-quality performance. An example of the proposed hybrid prompt engineering strategy is shown in Fig. 3.

3.2. Toolset module

This module converts design codes and formulas as logic tools and mathematical tools, respectively, for calculation decisions, formula selection, partial coefficient and settlement formulas. The logical tools are integrated as a coherent logic workflow using a series of “if-elif-else” statements. Upon receiving structured data from the LLM module, the logic workflow triggers corresponding mathematical tools for calculations and ultimately produce design solutions. For example, if a key-value pair is “Standard: Chinese”, the logic tools recognize that the value of “Standard” is “Chinese”, and then call the Chinese design codes for mathematical calculations. The toolset module serves as an indispensable component in GeoLLM, ensuring calculation efficiency and precision in geotechnical design.

4. Single pile case study

This section aims to validate the capacity of GeoLLM in understanding geotechnical texts and calculating the bearing capacity and settlement of single piles. The single pile cases involve the general procedure of geotechnical design, including type selection, code

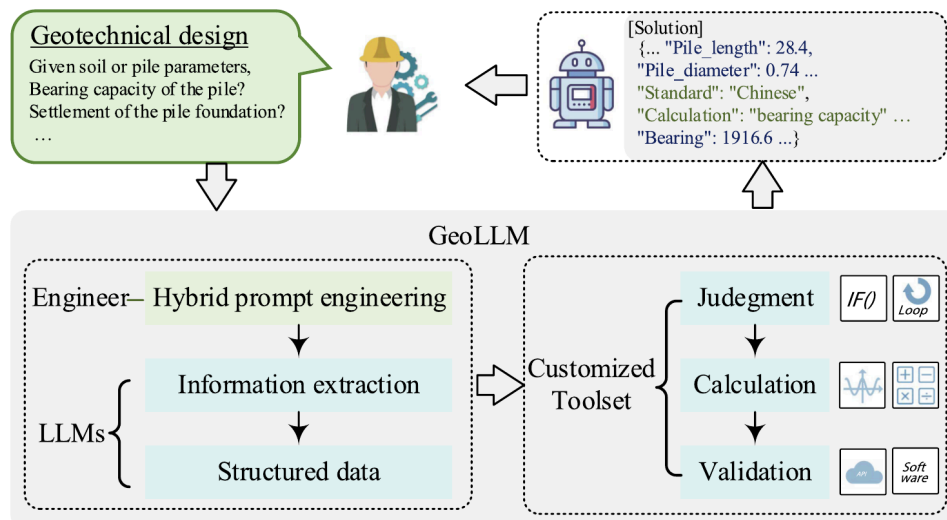


Fig. 2. GeoLLM framework for intelligent geotechnical design.

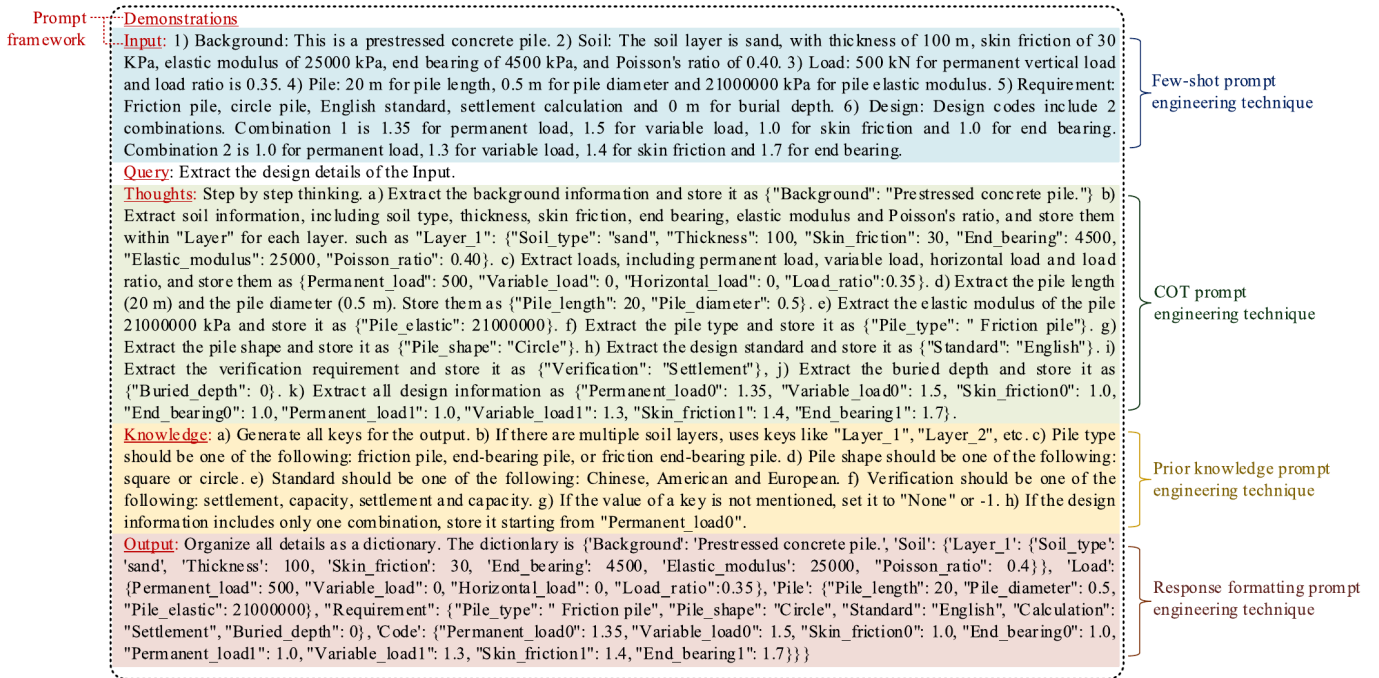


Fig. 3. Example of hybrid prompt engineering strategy in GeoLLM.

selection, bearing capacity calculation and settlement calculation. In this case study, 54 scenarios are evaluated, involving 3 pile types (friction, end bearing, friction & end bearing), 2 cross-sectional shapes (circle and square), 3 design codes (European (EN), Chinese (GB), American (USA)), and 3 calculation requirements (settlement, load bearing, both). These cases can facilitate a comprehensive assessment on the reliability and accuracy of LLMs in geotechnical design.

4.1. Test dataset

A single pile dataset comprising 162 cases is well-prepared to evaluate the reliability and accuracy of GeoLLM. These cases are divided into 3 sub-tests, involving 54 standard cases, 54 simplified cases, and 54 free cases. Each sub-test covers all 54 scenarios with varying semantic complexity through diverse expressions and language order, and soil distributions. Standard cases are sourced from the textbook of Das (2018), entitled "Principles of foundation engineering". This book provides comprehensive case studies on various design tasks and is used as the reference book in universities all over the world. Simplified cases involve random omission of specific information from standard cases for minor alterations, such as pile type, pile shape, design code, buried depth and design requirement. Free cases are generated by employing LLMs to significantly rearrange the word orders and expressions in standard cases, which are totally different from standard cases. 3 representative single pile cases are shown in Fig. 4.

4.2. GeoLLM modelling

The modelling procedure includes an LLM module for the extraction of structured data and a toolset module for the calculation of bearing capacity and settlement of single piles, as shown in Fig. 5.

4.2.1. Single pile LLM module

In this module, Gemini-pro (Anil et al., 2023) is adopted as the backbone. The generation of structured data is the process of receiving single pile texts and extracting key information as key-value pairs. The performance of extraction significantly affects text understanding and the accuracy of the subsequent mathematical calculations. This module

is divided into 2 sub-steps: step 1 for information organization and step 2 for structured data generation, as shown in Fig. 6. Information organization focuses on processing diverse texts and extracting useful details in a standardized format, including background, soil properties, load specifics, pile information and design requirement. In this sub-step, few-shot and response formatting techniques are leveraged to form the prompt P1. Structured data generation refers to generating structured data from useful details through a hybrid prompt P2 comprising of few-shot, prior knowledge, COT and response formatting techniques.

4.2.2. Mathematical toolset

The mathematical toolset includes the logic tools and mathematical tools for bearing capacity verification and settlement calculation, respectively.

(1) Bearing capacity verification involves iterative calculation of single pile bearing capacity according to design requirements. If the requirements are not satisfied, the toolset module will iteratively adjust the length or diameter of the pile, based on the ratio of load to bearing capacity and the safety factor defined in design codes. The Chinese codes use the safety factor method, while European and American codes use the partial factor method (Tang et al., 2023b). These methods can be easily integrated into GeoLLM by in-house functions and will be activated if bearing capacity verification is needed. Suitable mathematical tools are used to verify single pile bearing capacity, as shown in Fig. 7. F_l is the design load. r_p is the permanent load factor. F_p is the permanent load. r_v is the variable load factor, F_v is the variable load. R_e represents the end-bearing resistance. A_p is the cross-sectional area of piles. q_{st} denotes the end-bearing resistance per unit area. p is the perimeter of the piles. L_j indicates the embedded length in soil layer j . f_{sj} is the pile-around friction per unit area. R_s is the frictional resistance. R_p is the load bearing of the piles. r_{safe} is the safety factor. r_e is the end-bearing factor. r_s is the frictional factor. Their verification process for EU requires 2 times, so $\bullet_1$ and $\bullet_2$ are used as subscripts to uniformly distinguish them.

(2) Settlement calculation, the process of determining the settlement of a single pile under specific loads, will be activated if it is identified in the structured data. Single pile settlement is generally determined by the combined effects of average axial pressure, pile-tip pressure, and pile-

Simplified case

The context is outlined below. With the ongoing expansion of urban areas, there is a heightened need for large structures like skyscrapers and bridges. Addressing this need, geotechnical engineers are progressively adopting the use of individual piles as a foundation method due to its effectiveness. The arrangement of the soil layers is detailed below. The first soil layer is compacted fill, with thickness of 2 m, skin friction of 25 kPa, elastic modulus of 9×10^3 kPa, unknown end bearing, and Poisson's ratio of 0.40. The second layer is loose silt, with thickness of 3 m, skin friction of 20 kPa, elastic modulus of 16×10^3 kPa, unknown end bearing, and Poisson's ratio of 0.40. The third layer is stiff soil, with thickness of 6 m, skin friction of 35 kPa, elastic modulus of 23×10^3 kPa, unknown end bearing, and Poisson's ratio of 0.40. The fourth layer is silty sand, with thickness of 4 m, skin friction of 35 kPa, elastic modulus of 22×10^3 kPa, end bearing of 2500 kPa, and Poisson's ratio of 0.40. The fifth soil layer is gravel, with thickness of 50 m, skin friction of 95 kPa, elastic modulus of 36×10^3 kPa, end bearing of 4750 kPa, and Poisson's ratio of 0.40. Load information encompasses 957 kN for permanent vertical load, 347 kN for variable vertical load, 352 kN for horizontal load. The load ratio is 0.35. Pile information encompasses 19 m for pile length, 0.55 m for pile diameter, 21×10^6 kPa for pile elastic modulus. Verification requirements are not given. The design codes include 1.4 for permanent load, 1.4 for variable load, 2.1 for safety factor.

Note: Changed content

Remove verification requirement



Standard case

As urbanization continues to progress, there is a growing demand for large-scale structures such as high-rise buildings and bridges. To meet this demand, geotechnical engineers are increasingly utilizing single piles as an efficient foundation solution. The soil layer distribution is as follows. The first soil layer is compacted fill, with thickness of 2 m, skin friction of 25 kPa, elastic modulus of 9×10^3 kPa, unknown end bearing, and Poisson's ratio of 0.40. The second layer is loose silt, with thickness of 3 m, skin friction of 20 kPa, elastic modulus of 16×10^3 kPa, unknown end bearing, and Poisson's ratio of 0.40. The third layer is stiff soil, with thickness of 6 m, skin friction of 35 kPa, elastic modulus of 23×10^3 kPa, unknown end bearing, and Poisson's ratio of 0.40. The fourth layer is silty sand, with thickness of 4 m, skin friction of 35 kPa, elastic modulus of 22×10^3 kPa, end bearing of 2500 kPa, and Poisson's ratio of 0.40. The fifth soil layer is gravel, with thickness of 50 m, skin friction of 95 kPa, elastic modulus of 36×10^3 kPa, end bearing of 4750 kPa, and Poisson's ratio of 0.40. Load information includes 957 kN for permanent vertical load, 347 kN for variable vertical load, 352 kN for horizontal load. The load ratio is 0.35. Pile information includes 20.5 m for pile length, 0.45 m for pile diameter, 22×10^6 kPa for pile elastic modulus. Verification requirements include friction end-bearing pile, square pile, American standard, settlement and bearing capacity calculation, 0 m for burial depth. The design codes include 1.4 for permanent load, 1.4 for variable load, 1.5 for skin friction, 1.5 for end bearing.

Change language Expression and mode



Free case

Geotechnical Profile 1. Compacted Fill: - Thickness: 2 m - Skin Friction: 25 kPa - Elastic Modulus: 9×10^3 kPa - Poisson's Ratio: 0.40 2. Loose Silt: - Thickness: 3 m - Skin Friction: 20 kPa - Elastic Modulus: 16×10^3 kPa - Poisson's Ratio: 0.40 3. Stiff Soil: - Thickness: 6 m - Skin Friction: 35 kPa - Elastic Modulus: 23×10^3 kPa - Poisson's Ratio: 0.40	4. Silty Sand: - Thickness: 4 m - Skin Friction: 35 kPa - Elastic Modulus: 22×10^3 kPa - End Bearing: 2500 kPa - Poisson's Ratio: 0.40 5. Gravel: - Thickness: 50 m - Skin Friction: 95 kPa - Elastic Modulus: 36×10^3 kPa - End Bearing: 4750 kPa - Poisson's Ratio: 0.40 6. Load Information: - Permanent Vertical Load: 957 kN - Variable Vertical Load: 347 kN - Horizontal Load: 352 kN - Load Ratio: 0.35	7. Pile Information: - Pile Length: 20.5 m - Pile Diameter: 0.45 m - Pile Elastic Modulus: 22×10^6 kPa 8. Verification Requirements: - Pile Type: Friction End-Bearing Pile - Pile Shape: Square Pile - Design Code: American Standard - Calculations: Settlement & Bearing Capacity - Burial Depth: 0 m 9. Design Codes: - Permanent Load Factor: 1.4 - Variable Load Factor: 1.4 - Skin Friction Factor: 1.5 - End Bearing Factor: 1.5
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Fig. 4. Examples of standard, simplified, and free cases in single pile dataset.

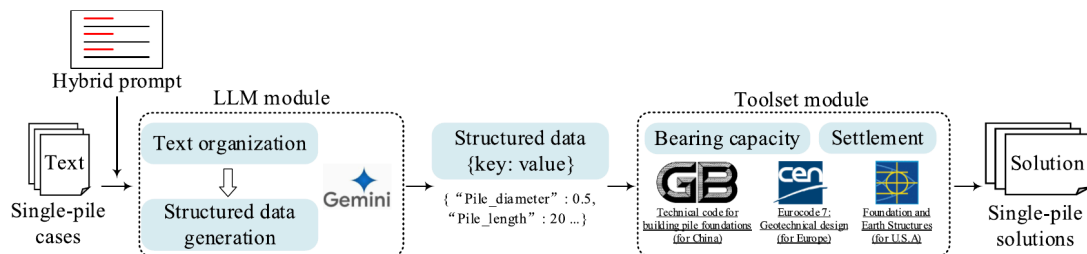


Fig. 5. Flowchart of GeoLLM for single pile case.

Text input of a single-pile case

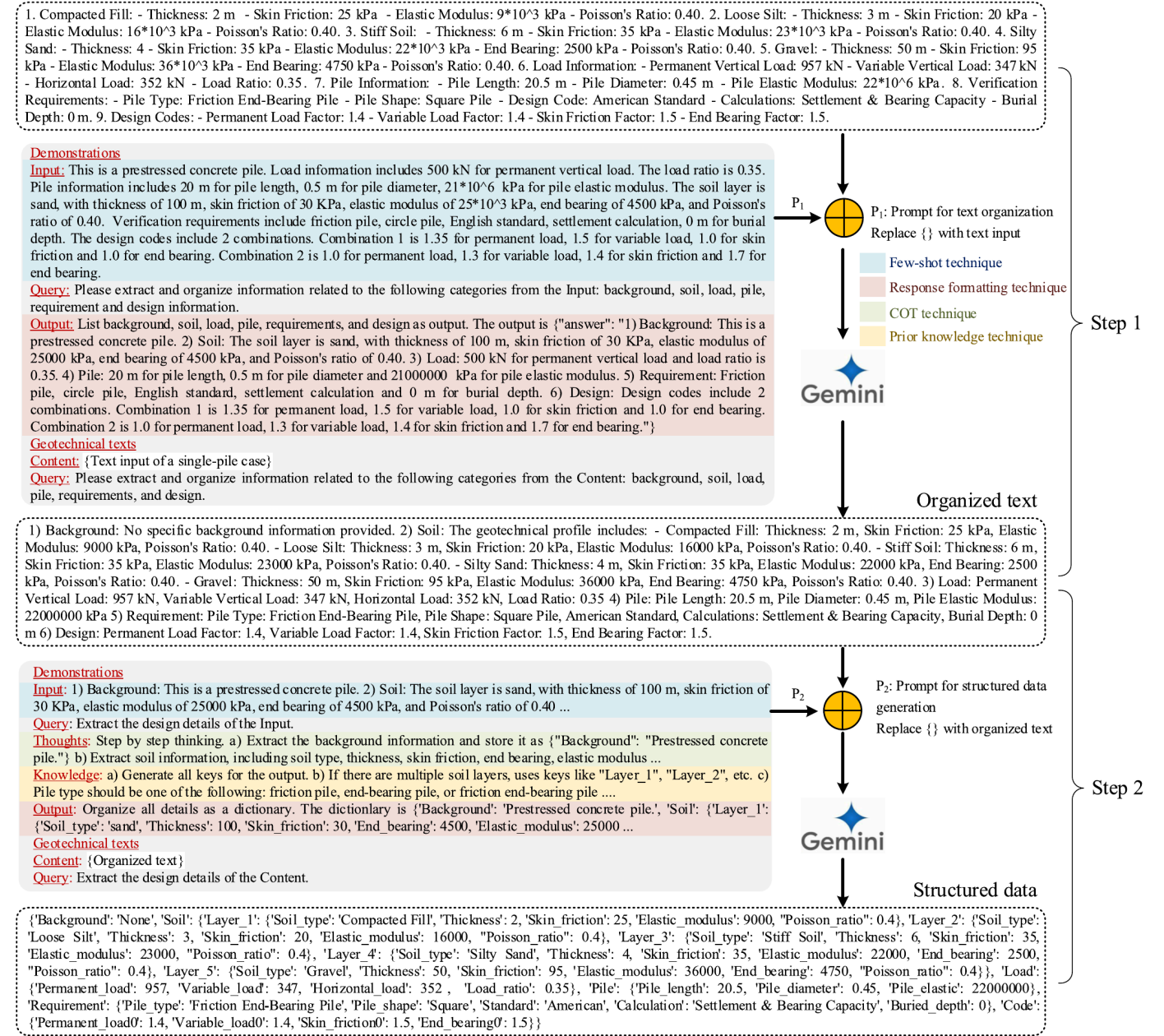


Fig. 6. Single pile LLM module for generating structured data.

around friction (Das, 2018). Average axial pressure induces axial compression settlement $S_{e(1)}$, calculated by Eq. (1). Pile-tip pressure causes end-bearing settlement $S_{e(2)}$, calculated by Eq. (2). Pile-around friction results in friction settlement $S_{e(3)}$, calculated by Eq. (3). The total settlement of a single pile is the sum S_E of the settlement $S_{e(1)}$, $S_{e(2)}$ and $S_{e(3)}$.

$$S_{e(1)} = \frac{(Q_{wp} + \xi Q_{ws})L}{A_p E_p} \quad (1)$$

where Q_{wp} is the load carried at the pile tip. Q_{ws} is the load carried by frictional resistance. E_p is the elasticity modulus of a pile. L is the pile length. ξ is load factor, as 0.5.

$$S_{e(2)} = \frac{q_{wp} D}{E_{st}} (1 - \mu_{st}^2) I_{wp} \quad (2)$$

where D is the width or diameter of a pile. q_{wp} is the point load per unit

area at the pile tip, as Q_{wp}/A_p . E_{st} is the elasticity modulus of the soil under the pile tip. μ_{st} is the Poisson's ratio of the soil under the pile tip. I_{wp} is the influence factor of the soil under the pile tip, as 0.85.

$$S_{e(3)} = \sum_{j \in \text{fr.}} \left(\frac{Q_{ws}}{p L_j} \right) \frac{D}{E_{sj}} (1 - \mu_{sj}^2) I_{wsj} \quad (3)$$

where E_{sj} is the elasticity modulus of soil layer j . μ_{sj} is the Poisson's ratio of soil layer j . I_{wsj} is the influence factor of soil layer j , as $2 + 0.35 \sqrt{L_j/D}$.

4.3. Evaluation metrics

The evaluation metrics for assessing the performance of GeoLLM involve accuracy, precision and recall (Powers, 2011). Accuracy is used to evaluate final solutions, while precision and recall are used to evaluate structured data. All these metrics range between 0 and 1. In this study, 162 single pile cases are imported to LLM module to generate

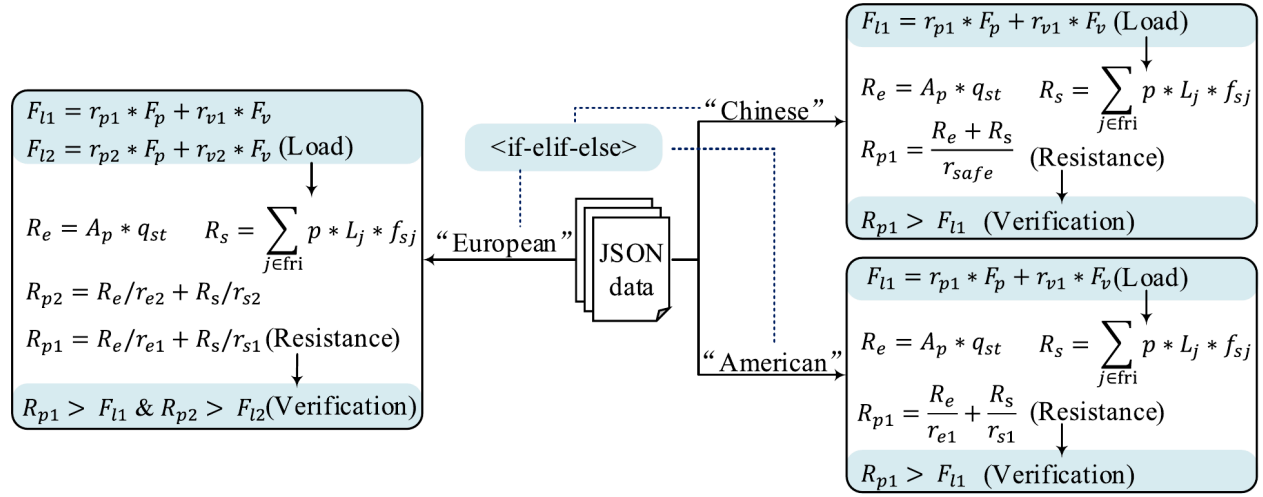


Fig. 7. Mathematical and logic tools for bearing capacity calculation.

corresponding structured data and calculate the required solutions (the bearing capacity and settlement) using toolset module as shown in Fig. 5, with each case being tested over 3 runs. For each case, GeoLLM will activate the LLM module to generate structured data and terminate the connection once the process is complete, avoiding the influence of history queries. Accuracy refers to the proportion of correct solutions over the total runs (162×3). The toolset module can calculate the correct solution only when all the key-value pairs are accurately extracted in structured data. Therefore, the accuracy is equal to the proportion of cases where all key-value pairs are extracted accurately to the total runs. Precision P_i and recall R_i are utilized to assess the quality of structured data extracted by the GeoLLM. Precision P_i refers to average ratio of true key-value pairs to total pairs within generated structured data by GeoLLM for each sub-test, as shown in Eq. (4). It represents the accuracy of GeoLLM to extract true key-value pairs. Recall R_i refers to average ratio of true key-value pairs generated by GeoLLM to the total pairs within the ground-truth structured data for each case, as shown in Eq. (5). It specifically represents the completeness of key-value pairs extracted by GeoLLM in comparison to the ground-truth pairs.

$$P_i = \frac{1}{N_c} \sum_{i=1}^{N_c} \frac{TP_i}{FP_i + TP_i} \quad (4)$$

where N_c is the number of single pile cases, set as 54 for standard, word-level, and free sub-tests, respectively. TP_i is the number of true key-value pairs generated by GeoLLM for the i^{th} single pile case over 3 runs. FP_i is the number of false key-value pairs generated by GeoLLM for the i^{th} single pile case over 3 runs. TP_i plus FP_i equals the number of total key-value pairs in generated structured data for the i^{th} single pile case over 3 runs.

$$R_i = \frac{1}{N_c} \sum_{i=1}^{N_c} \frac{TP_i}{FN_i + TP_i} \quad (5)$$

where FN_i is the number of true key-value pairs not generated by

	Key-value pair in ground-truth data	Key-value pair NOT in ground-truth data
Key-value pair in generated data	True positive (TP)	False positive (FP)
Key-value pair NOT in generated data	False negative (FN)	True negative (TN)

Fig. 8. Confusion matrix of TP_i , FN_i and FP_i .

GeoLLM for the i^{th} single pile case over 3 runs. TP_i plus FN_i equals the number of total key-value pairs in ground-truth structured data corresponding to the i^{th} single pile case over 3 runs. Fig. 8 uses a key-value pair as an example to further explain TP_i , FP_i plus FN_i in Eq. (4) and (5).

5. Results and discussions

The performance of GeoLLM in single pile cases are comprehensively analysed and discussed in this section, in terms of performance, effect of LLM architecture and size, computational cost, limitations and future work.

5.1. Performance on single pile case

Table 3 tabulates the accuracy, precision and recall of GeoLLM built with Gemini-pro in 3 sub-tests. In the standard sub-test, GeoLLM exhibited high levels of accuracy (0.988), precision (0.995), and recall (0.997). The simplified sub-test results showcased the robust performance, particularly in terms of precision (0.994) and recall (0.995), despite a slight decrease in accuracy (0.932). The free sub-test achieved satisfactory results with accuracy, precision, and recall all achieving 0.92.

The accuracy result showcases the remarkable effectiveness of GeoLLM in processing geotechnical texts closely aligning with the prompts. When the geotechnical text deviates from few-shot examples within prompts, the risk of errors significantly increases. Regarding to free sub-test, GeoLLM consistently demonstrates an impressive level of proficiency, achieving a commendable accuracy rate of 0.92. High precision and recall indicate the robust performance of GeoLLM in generating true key-value pairs.

Fig. 9 illustrates the distribution of precision and recall across three sub-tests through box and violin plots. In the standard sub-test, the precision and recall are consistently clustered around 1 in box plots with tails at 0.95 and 0.97 in the violin plots, showing no extreme cases. For the simplified sub-test, precision demonstrates commendable performance with tails at 0.94, while recall exhibits minor fluctuations with tails at 0.96 compared to the standard sub-test. In the free sub-test,

Table 3
Evaluation metrics of GeoLLM in single pile dataset.

Sub-test	Accuracy	Precision	Recall
Standard sub-test	0.988	0.995	0.997
Simplified sub-test	0.932	0.994	0.995
Free sub-test	0.920	0.920	0.919

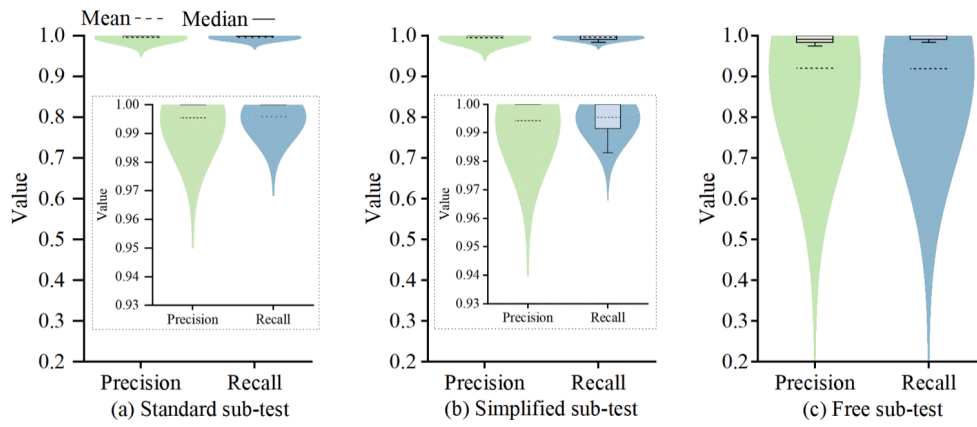


Fig. 9. Performance on structured data extracted by GeoLLM in different sub-tests. (a) Standard sub-test, (b) Simplified sub-test and (c) Free sub-test.

although most cases maintain high precision and recall values (over 0.97), a few extreme cases display significant deviations with tails at 0.2. The substantial decline in performance for free cases stems from the large discrepancies between these cases and prompt examples. As the discrepancies increase, the effectiveness of the prompts in guiding LLMs diminishes. This reduction leads to an increase in error rates for GeoLLM, making it more susceptible to failure. Standard sub-test is more stable compared to simplified sub-test, which in turn is more stable than free sub-test.

The results encompass 18 failures in settlement-related cases and 20 failures in bearing capacity-related cases. All errors related to key-value pairs are counted as shown in Fig. 10. The term “Wrong background” refers to instances where the value of the background key contains errors or misses some information, while “No-recalled background” pertains to situations where background information is lost during text processing, leading to a value of “None”. Similarly, “Wrong other pairs” describes instances where incorrect information is extracted, such as errors in pile type or calculation standards, and “No-recalled other pairs” indicates that certain keys, like the elastic modulus of the pile, are missing.

Most errors in key-value pair are associated with the long background information. When dealing with lengthy geotechnical texts, LLMs are prone to extracting incomplete or omitted background

information, leading to inaccuracies in key-value pairs. This finding suggests potential deficiencies of GeoLLM in grasping complex geotechnical concepts from the lengthy background information. In addition, the errors in other pairs, such as pile type and calculation standards, seems to be randomly distributed, indicating an underlying challenge stemming from the lack of pre-training data for LLMs. This problem can be significantly mitigated through fine-tuning using geotechnical data or more sophisticated prompt engineering.

The capabilities of original LLMs are typically evaluated using single pile cases without the aid of prompts or external toolsets. It was found that original LLMs struggled in the calculations of bearing capacity and settlement, resulting in incorrect solutions with the accuracy of 0. Furthermore, in the absence of prompts for guidance, LLMs generated incorrect structured data, resulting in both precision and recall of 0. Lack of prompts lead to a significant decline across all evaluation metrics.

Among these metrics, high precision and recall scores collectively demonstrate that with well-designed prompts, LLMs can accurately extract geotechnical information as structured data. The high accuracy score reflects the robust capacity of computing geotechnical solutions. The observed decline in performance underscores the limited capability of the original LLMs to process geotechnical design tasks in the absence of appropriate prompts and engineering toolsets. Without proper prompts to guide the generation process, original LLMs tend to extract incorrect information and generate vast amounts of random and irrelevant data. Moreover, without toolsets for calculation and decision-making, solutions generated by original LLMs are always erroneous. These findings further substantiate the significance of a proper prompt engineering strategy and mathematical toolsets within the domain-specific GeoLLM framework.

5.2. Effects of different LLMs and size

In this section, eight representative LLMs have been compared and validated using standard cases, as shown in Table 4. For the size effect, the models, including Qwen-1.8B, Qwen-7B, Qwen-14B, Qwen-72B, and Qwen-plus (Bai et al., 2023) were chosen. Qwen is a family of large language models, available on Hugging Face. To evaluate the architecture effect, Qwen-plus, GPT-4 (Achiam et al., 2023), GLM-4 (Zeng et al., 2023) and Gemini-pro (Anil et al., 2023) were utilized.

Fig. 11 illustrates the accuracy, precision and recall among LLMs with different sizes. The results exhibit a significant increase in accuracy rising from 0 % to 97.5 % as the parameters increase from 1.8B to 100B in LLMs. Similarly, other metrics also present a similar phenomenon, with initial values of 0.9 % (precision) and 7.1 % (recall) for model with 1.8B parameters, increasing to 98.9 % (precision) and 99.5 % (recall) for model over 100B parameters.

This figure reveals that as the size of the LLMs increases, the

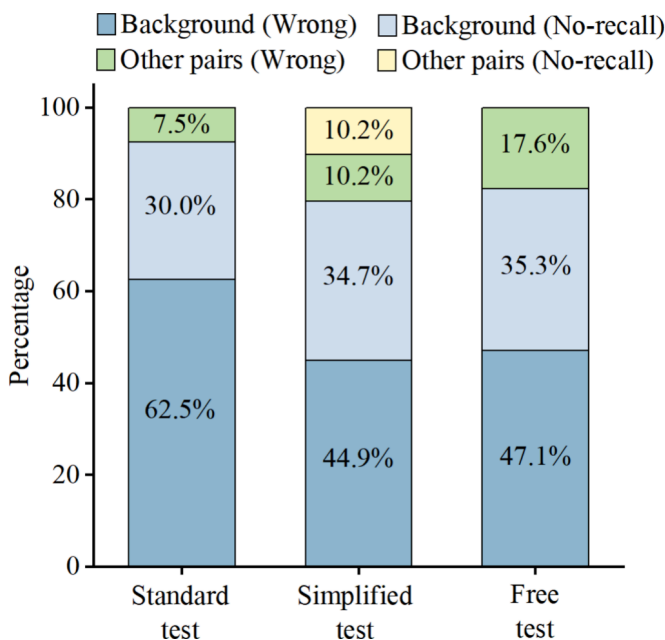


Fig. 10. Distribution of key-value pair errors in single pile dataset.

Table 4
Characteristics and computational effort of representative LLMs.

Specific LLM	Parameter magnitude (bits)	Deployment (Nvidia RTX 4090)	Deployment requirement	Inference time	Cost (750,000 words)
Qwen-1.8B	6B (INT 4)	local PC	3 GB memory	49.05 s	0 \$ for input & output
Qwen-7B	7B (INT 4)	local PC	9 GB memory	239.37 s	0 \$ for input & output
Qwen-14B	14B (INT 4)	local PC	13 GB memory	505.24 s	0 \$ for input & output
Qwen-72B	72B	API	-	36.46 s	2.8 \$ for input & output
Qwen-plus	> 100B	API	-	38.05 s	0.6 \$ for input 1.7 \$ for output
GPT-4	> 100B	API	-	57.14 s	5 \$ for input 15 \$ for output
GLM-4	> 100B	API	-	50.22 s	14.3 \$ for input & output
Gemini-pro	> 100B	API	-	24.63 s	0.35 \$ for input 1.05 \$ for output

performance of GeoLLM exhibits significant improvement across all metrics. LLMs with over 100B parameters demonstrate exceptional performance, which are very suitable as the backbone of GeoLLM. LLMs with 1.8B or 7B can be enhanced by distillation techniques to learn insights from LLMs over 100B. If the construction of GeoLLM relies on LLMs with 14B or 72B parameters, it is crucial for engineers to fine-tune these models using high-quality geotechnical data. Notably, the 72B LLM exhibits obvious enhancement in accuracy compared with the 14B LLM, while the improvement in other metrics is relatively slight. This discrepancy is primarily due to the 72B model producing partially inconsistent-structure outputs after information organization. Despite their information being correct, these inconsistent structures always lead to information loss in the following processing steps. This loss impacts the overall evaluation of the 72B model. Thus, it is crucial to adapt and optimize the data format within GeoLLM to fully unlock its potential. The performance across all metrics of 72B LLMs is expected to significantly elevate by an optimized data format, surpassing 14B LLMs.

Fig. 12 demonstrates the performance of different types of LLMs based on 3 metrics. Notably, GPT-4 and GLM-4 achieve perfect accuracy

of 100 %, while Gemini-pro and Qwen-plus also perform commendably, trailing slightly with accuracy scores of 98.8 % and 97.5 %, respectively. Other metrics are all above 97 % for all these LLMs, where both GPT-4 and GLM-4 still slightly outperform Gemini-pro and Qwen-plus.

Across all evaluation metrics, these LLMs demonstrate exceptional performance, particularly GPT-4 and GLM-4 and their near-perfect scores. Although minor differences exist between various LLMs in terms of specific metrics, all these LLMs are suitable for building GeoLLM. When selecting LLMs with over 100B parameters for GeoLLM, computational effort plays a crucial role. Therefore, it is advisable to consult an effort analysis, as shown in Table 4, to guide decision-making.

5.3. Computational effort of GeoLLM

The computational costs are related to deployment types and the tokens of input and output. For local deployment, there are no computational costs involved. For cloud deployment using APIs, the computation costs are tied to the cloud service providers and are typically billed based on tokens. The computational costs are tabulated in Table 4. These results indicate that as the size of the LLM increases, the inference time of GeoLLM surges rapidly. In terms of efficiency, constructing GeoLLM using APIs is notably quicker than local deployment, but it also results in additional costs for cloud service.

5.4. Limitations and future work of GeoLLM

While GeoLLM holds immense potential, it is not without limitations.

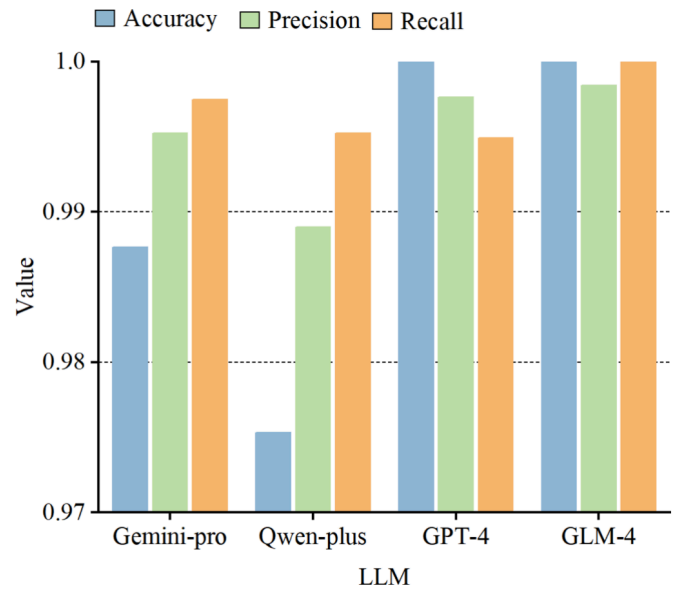


Fig. 12. Comparative performance of LLMs with different architectures.

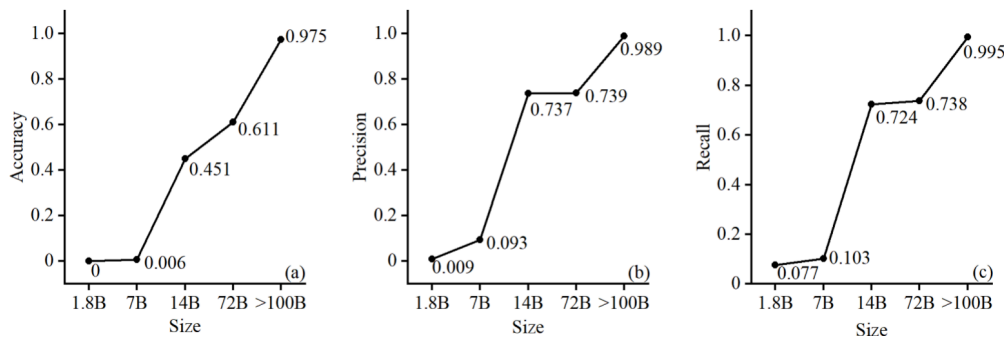


Fig. 11. Evaluation metrics of Qwen models with different sizes. (a) Accuracy, (b) Precision and (c) Recall.

For example, GeoLLM may generate incorrect responses occasionally, and cannot self-identify and correct these errors as tabulated in Table 3, resulting a decrease in evaluation metrics. It is important to acknowledge that even LLMs with over 100 billion parameters cannot consistently guarantee 100 % accuracy in handling structured data and generating solutions. In addition, the use of unexplainable networks within GeoLLM limits its application in geotechnical design. Engineers prefer a white-box approach that reveals its operational principles. Moreover, engineers should carefully choose the suitable LLMs to assist them in geotechnical design considering accuracy, deployment conditions and computational effort. Therefore, further efforts are necessary to achieve a reliable and robust GeoLLM in the future.

Enhancing the capacities of GeoLLM requires self-review loops, enhanced LLMs, and flexible modular design. Self-review loops use established geotechnical principles with LLMs to enhance GeoLLM's capability to review and correct wrong responses. Enhanced LLMs not only provide more accurate and stable responses but also enable automatic planning and tool calling, eliminating the need for the traditional "if-elif-else" structure. This design can significantly enhance GeoLLM's application across various design codes. Modular design divides GeoLLM into diverse components, such as searching, drawing, calculation and auto-planning components. Engineers can select the components most relevant to their geotechnical designs to construct a tailored GeoLLM.

6. Conclusion

This study has developed a ground-breaking geotechnical AI framework GeoLLM that integrates LLMs and mathematical toolsets for intelligent geotechnical design, validated on the single pile design. Additionally, an innovative prompt engineering strategy for geotechnical design and a single pile dataset for evaluation were provided. Extensive tests on the single pile dataset revealed that GeoLLM performed exceptionally well, achieving an accuracy of 0.988 in the standard test, 0.932 in the simplified test, and 0.920 in the free test. Furthermore, the precision and recall were also outstanding, ranging from 0.919 to 1. The high metrics indicate the feasibility of GeoLLMs in comprehending geotechnical knowledge, suggesting the potential of GeoLLM in facilitating the intelligent geotechnical design.

The effect of LLM size was also explored. LLMs with 1.8B and 7B parameters struggled to understand any geotechnical text. Similarly, LLMs with 14B and 72B parameters exhibited comparatively lower performance. In contrast, LLMs with over 100 billion parameters exhibited outstanding comprehensive capabilities. As the size of LLMs increased, the capacities of LLMs in understanding geotechnical texts, processing effective information, and generating accurate responses are enhanced. Accordingly, engineers can select appropriate LLMs to build GeoLLM based on their requirements and deployment conditions, such as under-14B LLMs with distillation techniques, 14B-72B LLMs with fine-tuning or over-100B LLMs. The evaluation results of different LLMs, such as Gemini, Qwen, GPT, and GLM, revealed subtle differences in accuracy, precision, and recall scores. However, these LLMs with over 100B parameters consistently achieved high-quality results. Thus, engineers have the option to suitable such as GPT, GLM, Gemini and Qwen.

GeoLLM also possesses 3 potential limitations, including the lack of self-correction, explainability, and deployment requirement. Therefore, the adoption of self-review loops, enhanced LLMs and modular design to enhance its application in geotechnical design is planned in future work.

CRedit authorship contribution statement

Hao-Ruo Xu: Writing – original draft, Validation, Software, Methodology. **Ning Zhang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Zhen-Yu Yin:** Writing – review & editing, Supervision, Funding acquisition. **Pierre Guy Atangana Njock:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

The research data, including the single pile dataset, chat histories, evaluation about this dataset, and a complete sample, can be found at <https://data.mendeley.com/preview/gx6k8xscbf>

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